
Penultimate-Layer Feature-Based OOD Detection: Feature Shaping and Its Limits

Martin Simons

Otto-Friedrich University of Bamberg
Aufbaustr. 11, 96049 Bamberg, Germany
martin.simons@stud.uni-bamberg.de

Seminar: xAI-Sem-M
Degree: M.Sc. (CitH)
Matriculation #: 2188932

Abstract

Post-hoc out-of-distribution (OOD) detection equips a frozen, pre-trained classifier to flag inputs unlike its training data, without retraining. We examine five feature-based methods that read the penultimate-layer representation: four feature-shaping rules—ReAct, ASH, VRA, and Optimal-FS—that transform the feature vector before rescoring, and BLOOD, which instead measures the smoothness of the network’s between-layer transformation. Although the methods share a layer and, in their mathematical form, assume no particular architecture, they were developed on different modalities and backbones and never co-evaluated under one protocol. Reading them together, we find three things: the shaping rules succeed only on the architecture they were designed on, with the data-optimized Optimal-FS and the conservative ReAct transferring while the aggressive pruning rules collapse; the cluster unifies only partially, as whole-vector ASH and the Jacobian-based BLOOD lie outside the optimization that explains the element-wise rules; and BLOOD, used as a control, shows the OOD signal is not carried by the feature values alone. A distance baseline, Mahalanobis, bounds both settings from above.

1 Introduction

Out-of-distribution (OOD) detection—recognizing when an input falls outside the distribution a model was trained on—has become a basic requirement for deploying machine-learning systems in the open world, where the inputs met at test time cannot all be anticipated during training and a confidently wrong prediction on an unfamiliar input can be costly. The need is sharpened by a shift in how models are built: practitioners increasingly rely on large pre-trained models they did not train and cannot easily retrain, using them as frozen checkpoints. This makes *post-hoc* OOD detection—methods that attach to an already-trained network at inference time, add no training cost, and leave its weights untouched—the dominant practical setting, and the one this report studies.

Within this setting, *feature-based* detectors read the penultimate-layer representation—the feature vector a network produces just before its final linear classifier—rather than the output logits alone. Most form a tight family this report calls the *shaping cluster*: ReAct [5] (rectified activations), ASH [1] (activation shaping), VRA [6] (variational rectified activation), and Optimal-FS [7]¹ each insert a fixed transformation that rewrites the feature vector before the logits are recomputed, sharpening the separation between in-distribution (ID) and OOD inputs while leaving ID classification essentially

¹Our shorthand for the otherwise unnamed method of Zhao et al. [7], after the paper’s title *Towards Optimal Feature-Shaping Methods*.

unchanged. BLOOD [2] (between-layer out-of-distribution detection) offers an alternative view of the same representation: rather than reshaping the feature vector, it measures how smoothly the network transforms it, on the premise that a trained model processes ID inputs more smoothly than OOD ones. Nothing in the mathematical setup of these methods refers to a particular architecture—each is defined for any frozen classifier with a penultimate layer—so on paper none is restricted to, or excluded from, any model family.

This report reads the five methods together. The shaping cluster rests on a shared assumption—that the discriminative OOD signal resides in the penultimate feature vector and can be recovered by reshaping it—and BLOOD, by reading a property of the transformation rather than editing the vector, serves as a control on that assumption. Because the methods were developed on different modalities and backbones and never co-evaluated under a single protocol, the report compares them with that incomparability in mind, asking three questions in turn: why the same method succeeds on some architectures and fails on others, how far the shaping cluster can be unified, and whether the OOD signal lies in the feature values or in the transformation applied to them.

2 Related Work

2.1 Positioning within the Survey Taxonomy

We situate the methods in the mechanism-oriented taxonomy of Lu et al. [4]. Its first cut separates *training-driven* methods, which alter the objective or the training data, from *training-agnostic* ones that operate on a frozen model; a second cut separates *test-time adaptive* methods, which update running statistics from test batches, from *post-hoc* methods, which score each input from a single forward pass. All five methods studied here are training-agnostic and post-hoc—the lightweight setting that dominates practice.

Post-hoc detectors differ in which model-internal quantity they read: the logits (e.g. MSP, MaxLogit, energy), a feature-space distance (e.g. Mahalanobis), a gradient, the penultimate representation, or an estimated density. This report concentrates on the *feature-based* family, which reads the penultimate activation vector $\mathbf{z} \in \mathbb{R}^M$ produced before the linear head $\mathbf{W} \in \mathbb{R}^{M \times C}$. Within it, Lu et al. [4] separate methods that reshape $\mathbf{z}$ directly from those that read a statistical property of it. The first branch—*feature shaping*—replaces $\mathbf{z}$ with $f(\mathbf{z})$ before the logits are recomputed, with f chosen to sharpen the ID/OOD gap while leaving ID accuracy intact; ReAct [5], ASH [1], VRA [6], and Optimal-FS [7] form a progression from hand-designed rules toward Optimal-FS’s optimization view, which recovers the earlier rules as approximations to a single ID-only optimum. BLOOD [2] sits in the second branch: it leaves $\mathbf{z}$ untouched and measures the smoothness of the network’s intermediate transformation through the feature Jacobian, and here serves as the control that isolates the shaping cluster’s shared assumption—that the OOD signal resides in $\mathbf{z}$ itself.

2.2 A Second Axis: Access Requirements

Beyond *which* quantity a method reads, Jelenić et al. [2] note a second, deployment-relevant axis—the *access* a method requires. *Black-box* methods need only inputs and outputs; *white-box* methods also need the weights but no data; *open-box* methods additionally need training data, an OOD validation set, or training intervention. This axis cuts across the cluster: ASH [1] and BLOOD [2] use no training data and are white-box, whereas ReAct [5], VRA [6], and Optimal-FS [7] fit thresholds or shaping functions to ID data and are open-box—so the sharp line is not BLOOD against the shaping cluster but the data-free methods (ASH, BLOOD) against the ID-data-dependent ones. The strongest open-box reference point, the Mahalanobis distance detector [3]—which fits a class-conditional Gaussian to ID features and scores by distance to the nearest class mean—belongs to neither paradigm; it appears only as a benchmark ceiling in the Optimal-FS and BLOOD experiments, to which we defer it.

3 Methodology

We consider a classifier pre-trained on in-distribution (ID) data and held frozen. For an input $\mathbf{x}$, the backbone produces a penultimate feature vector $\mathbf{z} \in \mathbb{R}^M$, which a linear head $\mathbf{W}$ maps to C class logits. A post-hoc detector reduces the network’s response to a scalar score $s(\mathbf{x}) \in \mathbb{R}$ and flags the

sample once that score crosses a threshold. Detection is thus a thresholding rule

$$G_\lambda : \mathbb{R} \rightarrow \{\text{ID}, \text{OOD}\}, \quad (1)$$

parameterized by a threshold λ , that labels $\mathbf{x}$ by comparing $s(\mathbf{x})$ to λ . The rule G_λ is common to every method; the methods differ only in how the score s is computed from the frozen network and in its orientation (which side of λ denotes ID). Detection quality is reported with the two standard metrics, AUROC (area under the receiver operating characteristic curve, a threshold-free measure of separability) and FPR95 (false positive rate at 95% true positive rate).

3.1 Feature Shaping

Feature shaping inserts a transformation $f : \mathbb{R}^M \rightarrow \mathbb{R}^M$ at the penultimate layer, reshaping $\mathbf{z}$ before the logits are recomputed and scored. The detector then factors into a shaping stage and a scoring stage,

$$f : \mathbb{R}^M \rightarrow \mathbb{R}^M, \quad s : \mathbb{R}^C \rightarrow \mathbb{R}, \quad s_{\text{shaped}}(\mathbf{x}) = s(\mathbf{W}f(\mathbf{z})), \quad (2)$$

and the sample is flagged by $G_\lambda(s_{\text{shaped}}(\mathbf{x}))$. The score s is a standard logit-based score (MSP, MaxLogit, or energy), for which a lower value indicates OOD; crucially, the choice of s is orthogonal to the shaping function f , a separation Optimal-FS later exploits. Setting $f = \text{id}$ recovers the unshaped baseline. Shaping functions are *element-wise* when f acts coordinatewise, $\bar{z}_i = f(z_i)$, and *whole-vector* when the transformation of one coordinate depends on the others; this distinction, carried entirely by the single symbol f , separates the methods below.

3.2 ReAct and VRA

ReAct [5] and VRA [6] are both element-wise. ReAct clips each activation at an upper threshold c ,

$$f_{\text{ReAct}}(z_i) = \min(z_i, c), \quad (3)$$

with c set to a high percentile (typically the 90th) of the activations estimated on ID data. The motivation is that OOD inputs tend to produce a few abnormally large activations; capping them suppresses the OOD score more than the ID score while leaving most ID activations, which fall below c , untouched.

VRA starts from a variational analysis of which operation maximally separates ID and OOD activations, concluding that the ideal rule suppresses both abnormally low and abnormally high activations. Its canonical piecewise realization, VRA-P, uses two ID-estimated quantile thresholds $\alpha < \beta$,

$$f_{\text{VRA-P}}(z_i) = \begin{cases} 0 & z_i < \alpha, \\ z_i & \alpha \leq z_i \leq \beta, \\ \beta & z_i > \beta. \end{cases} \quad (4)$$

This strictly generalizes ReAct, recovered as the special case $\alpha = 0$: VRA-P adds a lower cut that zeroes the abnormally small activations ReAct leaves in place. Both methods fit their thresholds to ID data and are therefore data-dependent, placing them in the open-box category (cf. Section 2.2). The amplification of mid-range activations (VRA+) and a feature-logit combined score (VRA++) are further refinements that depart from pure shaping; we summarize them in the appendix.

3.3 ASH

ASH [1] operates on the whole vector. For each sample it computes the p th percentile t of that sample’s own activations, prunes everything below t to zero, and then treats the surviving activations in one of three ways: ASH-P leaves them unchanged, ASH-B sets them all to a common constant, and ASH-S—the version we take as canonical—scales the survivors by a factor that restores the pre-pruning activation mass,

$$f_{\text{ASH-S}}(z_i) = \begin{cases} z_i \cdot \exp(s_1/s_2) & z_i \geq t, \\ 0 & z_i < t, \end{cases} \quad s_1 = \sum_j z_j, \quad s_2 = \sum_{j: z_j \geq t} z_j, \quad (5)$$

where s_1 and s_2 are the sums of activations before and after pruning. The premise is that the representation is heavily redundant, so most of it can be discarded and the sparse surviving support—rescaled to compensate—still classifies ID inputs while sharpening OOD separation. Because the

threshold t is computed per sample from the whole vector, ASH is whole-vector and not reducible to an element-wise map, and it uses no training statistics, making it data-free and therefore white-box (cf. Section 2.2). An extreme ablation, ASH-Rand, replaces the surviving values with random ones; we return to it in the discussion, as it bears on what ASH actually exploits.

3.4 Optimal-FS

Optimal-FS [7] recasts element-wise shaping as an optimization problem. Writing the per-coordinate reshaping as a multiplicative factor $\theta(z_i) = f(z_i)/z_i$, the shaped maximum logit is

$$\bar{\ell}_{\max}(\bar{\mathbf{z}}) = \sum_{i=1}^M \theta(z_i) w_i^{\max} z_i, \quad (6)$$

so the question becomes which feature *value ranges* should be up- or down-weighted. Optimal-FS partitions the feature range into K intervals, treats θ as piecewise constant over them, and collects the per-interval contributions into a K -dimensional interval-specific feature impact (ISFI) vector $\mathbf{I}(\mathbf{z})$, giving $\bar{\ell}_{\max} = \boldsymbol{\theta}^\top \mathbf{I}(\mathbf{z})$. The shaping vector $\boldsymbol{\theta}$ is then optimized to separate the ID and OOD maximum logits under a norm constraint; solved with OOD data it closely matches the element-wise rules above, recovering them as approximations to a common optimum. Dropping the OOD term, which contributes little beyond the ID direction, gives a closed-form solution from ID data alone (the optimization problem and its reduction are in Appendix A),

$$\boldsymbol{\theta}^* = \frac{\sqrt{K}}{\|\mathbb{E}_{\text{ID}}[\mathbf{I}(\mathbf{z})]\|} \mathbb{E}_{\text{ID}}[\mathbf{I}(\mathbf{z})]. \quad (7)$$

Nearly all of the benefit comes from $K \leq 10$, indicating the optimal shaping function has a simple form (the full ISFI derivation is in the appendix). Two caveats matter for this report. First, the optimization is specific to element-wise shaping, so it unifies ReAct and VRA-P but explicitly *does not* capture whole-vector methods such as ASH, which the authors name as an open problem. Second, like ReAct and VRA, it depends on ID data, making it open-box (cf. Section 2.2).

3.5 BLOOD

BLOOD [2] reads a different signal from the same region of the network. Rather than reshaping $\mathbf{z}$, it measures how smoothly the network transforms its representations, on the premise that a trained model maps ID inputs through smoother transformations than OOD ones. Smoothness at layer l is the squared Frobenius norm of that layer’s Jacobian $\mathbf{J}_l$,

$$\phi_l(\mathbf{x}) = \|\mathbf{J}_l(\mathbf{h}_l)\|_F^2, \quad (8)$$

estimated without forming the Jacobian through an unbiased random-probe estimator $\hat{\phi}_l$ (Appendix A). BLOOD uses this quantity directly as its score, $s(\mathbf{x}) = \hat{\phi}_{L-1}(\mathbf{x})$, with the *inverted* orientation relative to the shaping cluster: a higher value indicates OOD. The detector G_λ still applies, only with the sides of λ reversed. Note that this score is not expressible in the shaping composition $s \circ \mathbf{W} \circ f$ of Section 3.1: it reads the Jacobian and never factors through the head or a shaping map, which is precisely why BLOOD sits outside feature shaping. The method has two variants, BLOOD_M averaging $\hat{\phi}_l$ over all layers and BLOOD_L using only the final transformation, the Jacobian of the head with respect to the penultimate representation $\mathbf{z}$. The authors focus on BLOOD_L , which outperforms BLOOD_M more often, attributing this to the final layer undergoing the most training-driven change; this coincides with our penultimate-layer scope, so we take BLOOD_L as canonical. BLOOD requires only the model’s weights and no training data, making it white-box (cf. Section 2.2).

4 Experiments

The five methods were never evaluated under a shared protocol, and no honest unified leaderboard exists: they differ in modality (vision vs. text), backbone, OOD-dataset suite, and scoring function. Rather than force a comparison the literature does not support, we read the evidence as two internally consistent *islands*—vision feature shaping and text between-layer smoothness—and treat their incomparability as itself a finding: these methods target different deployment settings, and their relative merits are only defined within a setting. All figures below are reproduced from the cited papers, not independently re-run, and are comparable only within an island.

4.1 Vision: the shaping cluster under one protocol

Optimal-FS [7] provides the only head-to-head evaluation of the shaping cluster, re-running ReAct, ASH (P/B/S), VRA-P, and its own method across eight ImageNet-1k backbones under one protocol. On ResNet-50—the architecture all of these methods were designed on—the shaping methods are strong and close together (AUROC ≈ 90.3 for ASH-S, 88.6 for VRA-P, 86.5 for ReAct, 88.6 for Optimal-FS). The picture inverts off ResNet-50. On transformer and multi-layer-perceptron (MLP) backbones the most aggressive shaping rules collapse toward chance: ASH-S falls to 18.5 AUROC on ViT-B-16 (a Vision Transformer) and 33.9 on MLP-B, and VRA-P behaves similarly, dragging their eight-backbone averages to ≈ 38 . ReAct degrades far more gracefully (averaging 81.3, never falling below the high 70s), and Optimal-FS holds across the board (83.0 average, 82.7 on ViT-B-16). So the cluster’s apparent unity is conditional on its home architecture; the clip-based ReAct and the ID-optimized Optimal-FS generalize, while the percentile-prune methods do not. On the smaller-scale CIFAR benchmarks, feature shaping as a whole underperforms simple logit baselines, with Optimal-FS competitive but not dominant. Two caveats apply: these numbers come from a paper with a stake in the comparison, and the eight-dataset ImageNet averaging differs from the four-dataset suites used in the original ASH and ReAct papers.

4.2 Text: between-layer smoothness

BLOOD [2] evaluates a disjoint setting—text classification with RoBERTa and ELECTRA over eight ID datasets—so it neither confirms nor contradicts the vision results. Among methods requiring no training data (white/black-box), BLOOD_L is the strongest on most datasets, and more consistently than the all-layer variant BLOOD_M; the authors accordingly focus on BLOOD_L, attributing its edge to the final layer absorbing the most training-driven change. The advantage is more consistent on ELECTRA than on RoBERTa, with BLOOD_L significantly improving over the MSP baseline on six of the eight ELECTRA datasets [2]. Notably, the two shaping methods that appear here, ASH and ReAct, are only middling, indicating that vision-tuned shaping does not transfer to text. BLOOD’s grouping confirms our access axis (Section 2.2): its Table 1 splits the baselines into white/black-box and open-box columns, filing ASH as white-box but ReAct as *open-box*—explicitly because ReAct’s clip uses the 90th percentile of the *training-data* activations, the same data-dependence criterion our axis applies. BLOOD’s open-box set is therefore ReAct, ENSM (ensemble), TEMP (temperature scaling), and Mahalanobis, so the data-free/data-dependent line we draw coincides with BLOOD’s rather than cutting more strictly than it.

4.3 The ceiling, and what the islands show

A distance-based open-box method, Mahalanobis distance [3], bounds both islands from above. In BLOOD’s comparison it outperforms BLOOD in every setup but one (ELECTRA on TREC), and it is consistently the top scorer on the transformer text backbones. Its dominance is, however, setting-specific rather than universal: on ImageNet-scale ConvNets it is weak (AUROC ≈ 55 on ResNet-50 in VRA’s table [6]), where its class-conditional Gaussian assumption is harder to satisfy. The islands therefore deliver a structural message, not a numerical ranking: each paradigm wins only within its own setting, and a simple distance detector tops both wherever its distributional assumptions hold.

5 Discussion

Placing the five methods together raises three questions: why the same method succeeds on some architectures and fails on others (Section 5.1), how far the shaping cluster unifies (Section 5.2), and whether the OOD signal lies in the feature values or in the transformation applied to them (Section 5.3).

5.1 Architecture dependence

The transfer failures of Section 4.1 follow from the mechanisms, not from incidental tuning. Optimal-FS reports the pattern directly: for shaping methods, convolutional-network performance is negatively correlated with performance on other architectures [7]. ASH’s pruning assumes a redundant, sparse

representation—excess features of an over-parameterized network [1], a ReLU (rectified linear unit) convolutional premise—whereas BLOOD reads transformation smoothness in the upper layers of a transformer, where fine-tuning concentrates [2]. Each method encodes the architecture it was designed on, which is why neither transfers.

ReAct gives a partial cause. It attributes the abnormal OOD activations to batch-normalization statistics estimated on ID data and applied to OOD inputs, making batch normalization the exploited signal rather than an obstacle [5]. ReAct also helps under weight and group normalization [5], so the effect is not specific to batch normalization. The architecture dependence is therefore real but its root cause is unresolved across the cluster.

5.2 How far the shaping cluster unifies

Optimal-FS defines shaping as optimization over functions $f : \mathbb{R}^M \rightarrow \mathbb{R}^M$ and proves that ReAct and VRA-P, rewritten via $\theta(z) = f(z)/z$ (Eq. 6), approximate the optimal element-wise function [7]. This optimum is simple: the large jump above the baseline already appears at a single interval ($K = 1$), which prunes only the 0.1% tails of the feature distribution, and almost all remaining benefit is captured by $K \leq 10$ [7]. This explains why ReAct, a single clip, stays competitive with the optimization-derived method on the architectures both were tuned for: little structure beyond the tails remains to recover.

The unification is bounded, and Optimal-FS draws the boundary. Its tractable instance restricts to element-wise f and, in its own statement, fails to capture whole-vector methods such as ASH, which it still places inside the general $\mathbb{R}^M \rightarrow \mathbb{R}^M$ class [7]. ASH is thus excluded by the reduction, not the definition, and the whole-vector extension is named as open future work [7]. Extending this boundary to a case Optimal-FS does not treat, we observe that BLOOD lies outside the class: its score is the Frobenius norm of a Jacobian (Eq. 8) [2], not a map returning a reshaped feature vector, so no extension of the optimization can absorb it. This gives three tiers: ReAct and VRA-P inside the solved problem, ASH inside the general class but outside the solved instance, and BLOOD outside the class. The first gap is one of tractability; the second is categorical.

Whether the whole-vector extension is reachable is open. Two findings bear on it, and connecting them is our reading rather than a stated result. The element-wise optimum is coarse [7], and ASH’s signal is also coarse: ASH-Rand replaces the surviving activations with random values yet still beats the logit baselines, locating the signal in the surviving *support* rather than the value profile, while ASH-B and ASH-S—which discard the profile but preserve the activation mass (Eq. 5)—show the global scale is what that support must carry [1].

5.3 Where the signal lives

Because BLOOD is not a shaping function, it serves as a counterpoint that differs from the cluster on one variable: what is measured at the top of the network. The four shaping methods edit the activation vector; BLOOD leaves it unchanged and measures the smoothness of the transformation applied to it [2]. This poses a question neither paradigm sets out to answer: is the OOD signal a property of the feature values or of the map applied to them?

We do not resolve which is fundamental. The juxtaposition does narrow the cluster’s premise. BLOOD shows the activation vector is not the only carrier of OOD signal at the penultimate layer, and ASH-Rand shows that within the cluster the fine values are not the operative part. The shared assumption—that the signal can be recovered by reshaping the penultimate activation vector—is therefore narrower than its framing, though not incorrect.

6 Conclusion

We read five post-hoc, feature-based OOD detectors—the shaping cluster of ReAct, ASH, VRA, and Optimal-FS, together with BLOOD—as a single group united by the penultimate layer they all act on. Three conclusions follow. First, architecture, not mechanism, decides success: shaping rules tuned on ResNet-50 collapse on transformer and MLP backbones, and only the conservative clip of ReAct and the ID-optimized shaping of Optimal-FS transfer, so the cluster’s apparent unity is an artifact of its shared home architecture. Second, the unification is real but bounded: Optimal-FS’s optimization

recovers the element-wise rules as approximations to one ID-only optimum, but whole-vector ASH lies outside the solved instance and BLOOD, scoring a Jacobian norm, lies outside the shaping class entirely. Third, the OOD signal is narrower than the shaping premise suggests: BLOOD recovers signal without touching the feature values, and ASH-Rand recovers it without their fine structure. None of these methods is best in a setting-free sense; each encodes the architecture and modality it was born on, and a simple distance baseline outperforms both paradigms wherever its assumptions hold. We leave open whether the whole-vector case admits the same optimization account that unifies the element-wise rules.

References

- [1] A. Djurisić, N. Bozanić, A. Ashok, and R. Liu. Extremely simple activation shaping for out-of-distribution detection. In *International Conference on Learning Representations (ICLR)*, 2023. URL <http://arxiv.org/abs/2209.09858>.
- [2] F. Jelenić, J. Jukić, M. Tutek, M. Puljiz, and J. Šnajder. Out-of-distribution detection by leveraging between-layer transformation smoothness. In *International Conference on Learning Representations (ICLR)*, 2024. URL <http://arxiv.org/abs/2310.02832>.
- [3] K. Lee, K. Lee, H. Lee, and J. Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. *Advances in neural information processing systems*, 31, 2018.
- [4] S. Lu, Y. Wang, L. Sheng, L. He, A. Zheng, and J. Liang. Out-of-distribution detection: A task-oriented survey of recent advances. *ACM Computing Surveys*, 58(2):1–39, 2026. doi: 10.1145/3760390. URL <https://dl.acm.org/doi/10.1145/3760390>.
- [5] Y. Sun, C. Guo, and Y. Li. React: Out-of-distribution detection with rectified activations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. URL <http://arxiv.org/abs/2111.12797>.
- [6] M. Xu, Z. Lian, B. Liu, and J. Tao. VRA: Variational rectified activation for out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <http://arxiv.org/abs/2302.11716>.
- [7] Q. Zhao, M. Xu, K. Gupta, A. Asthana, L. Zheng, and S. Gould. Towards optimal feature-shaping methods for out-of-distribution detection. In *International Conference on Learning Representations (ICLR)*, 2024. URL <http://arxiv.org/abs/2402.00865>.

Use of Large Language Models

Declaration of Authorship

Ich erkläre hiermit gemäß § 9 Abs. 12 APO, dass ich die vorstehende Seminararbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Des Weiteren erkläre ich, dass die digitale Fassung der gedruckten Ausfertigung der Seminararbeit ausnahmslos in Inhalt und Wortlaut entspricht und zur Kenntnis genommen wurde, dass diese digitale Fassung einer durch Software unterstützten, anonymisierten Prüfung auf Plagiate unterzogen werden kann.

Bamberg, June 27, 2026

(Ort, Datum)

(Unterschrift)

A Derivations

Writing the per-interval contributions as the ISFI vector $\mathbf{I}(\mathbf{z})$ (Section 3.4), Optimal-FS optimizes the shaping vector $\boldsymbol{\theta}$ to separate the ID and OOD maximum logits under a norm constraint that prevents

trivial rescaling,

$$\max_{\boldsymbol{\theta}} \boldsymbol{\theta}^\top \left(\mathbb{E}_{\text{ID}}[\mathbf{I}(\mathbf{z})] - \mathbb{E}_{\text{OOD}}[\mathbf{I}(\mathbf{z})] \right), \quad \text{s.t. } \|\boldsymbol{\theta}\| = \sqrt{K}. \quad (9)$$

Dropping the OOD term, on the argument that it contributes little beyond the ID direction, leaves the closed-form $\boldsymbol{\theta}^*$ of Section 3.4.

For BLOOD, the layer- l smoothness $\phi_l = \|\mathbf{J}_l\|_F^2$ (Eq. 8) is estimated without forming the Jacobian, using N pairs of isotropic random probes $\mathbf{v}_i, \mathbf{w}_i$ (zero mean, unit variance),

$$\hat{\phi}_l(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N (\mathbf{w}_i^\top \mathbf{J}_l(\mathbf{h}_l) \mathbf{v}_i)^2, \quad \mathbb{E}[\hat{\phi}_l(\mathbf{x})] = \|\mathbf{J}_l\|_F^2, \quad (10)$$

the isotropy of the probes being what makes the estimator unbiased.

B Notation

Table 1: Unified notation used throughout this report.

Symbols are grouped by the work that introduces the corresponding quantity in the OOD-detection context. Typographic conventions—bold vectors and matrices, $\mathbb{1}\{\cdot\}$, and $\top$ for transpose—follow ?]. Where a symbol is renamed for cross-paper consistency, the original symbol is given in parentheses.

Symbol	Meaning
<i>General notation</i>	
x	Input sample
z	Penultimate-layer feature (activation) vector
z_i	i -th element of z
$\bar{z}$	Feature vector after shaping / rectification
M	Penultimate feature dimension
C	Number of classes
$\ell_i, \ell^{\max}$	i -th logit; max logit
$\mathbf{J}$	Jacobian matrix
$\ \cdot\ _F$	Frobenius norm
$\mathbb{1}(\cdot)$	Indicator (1 if condition holds, else 0)
$\mathbb{E}[\cdot]$	Expectation
<i>Sun et al. [5]</i>	
c	Activation clip threshold, $\text{ReAct}(z; c) = \min(z, c)$
$S(\cdot)$	OOD scoring function (MSP, MaxLogit, Energy, ...)
λ	OOD decision threshold
G_λ	OOD detector (in/out rule on S)
<i>Xu et al. [6]</i>	
$p_{\text{in}}, p_{\text{out}}$	ID / OOD marginal densities over activations
$g(\cdot), g^*(\cdot)$	Rectification function; variational optimum
τ	Fidelity-gap trade-off coefficient (orig. λ)
α, β	Low / high activation thresholds
γ	VRA+ amplification offset on the intermediate band
η_α, η_β	ID quantiles defining α, β
<i>?]</i>	
$\mathbf{w}^{\max}$	Weight vector of the predicted (max-logit) class
$\theta(\cdot)$	Element-wise shaping function, $\theta(z) := f(z)/z$
$\boldsymbol{\theta}$	Per-bucket shaping vector, $\boldsymbol{\theta} \in \mathbb{R}^K$
$I_{[a,b]}(\mathbf{z})$	Interval-specific feature impact (ISFI) of bucket $[a, b)$
a_k, b_k	Edges of the k -th bucket

continued on next page

Symbol	Meaning
K	Number of buckets ($K = 100$)
Jelenić et al. [2]	
$\phi_l(x)$	Score: squared Frobenius norm of the layer- l Jacobian
h_l	Layer- l representation ($h_{L-1} = \mathbf{z}$)
N	Number of random probes (orig. M)
$\mathbf{v}_i, \mathbf{w}_i$	Gaussian probe / projection vectors
$\mathbf{u}_i$	Jacobian-vector product, $\mathbf{u}_i = \mathbf{J}_{L-1} \mathbf{v}_i$
Djurisic et al. [1]	
p	Pruning percentile
t	Per-sample threshold, $t = \text{Perc}_p(\mathbf{z})$
m_i	Binary keep-mask, $m_i = \mathbb{1}\{z_i \geq t\}$
n	Number of surviving activations
s_1, s_2	Pre- / post-pruning activation sums